

Unit: 6 Unit 5: Linear Equations

Name: _____

Topic: Point-Slope Form, Slope-Intercept Form, Standard Form and Converting Forms

Date: _____

Challenge (Habanero)

Open Ended Questions (FRQ)

Q9. The amount of data y (in GB) used by players of an online game over time x (in hours) is modeled by the equation $y = 3x + 2$.

Graph the equation and interpret the slope and y -intercept in this context. Provide your response in [answer] format.

Q10. A game development company has a budget of 150,000 to develop a new game. The equation $y = -1500x + 150000$ represents the remaining budget in dollars after x hours of development. The equation is in slope form, given $(x_1, y_1) = (5, 120000)$. Then, interpret the slope and y -intercept in this context. Provide your response in

[answer] format.

Chef's Tips

1

- Tip 1: Ensure students understand the context of each question.
- Tip 2: Encourage students to read each question carefully and identify the relevant information.
- Tip 3: Remind students to show their work and provide explanations where necessary.